

C*-algebras and unitary representations of locally compact groups are deeply linked through functional analysis, operator algebras, and harmonic analysis. Here's a structured summary and study path to help you:

****1. The Link Between C*-Algebras and Unitary Representations**** C*-algebras naturally arise in the study of unitary representations of locally compact groups due to the following reasons:

1. ****Group C*-Algebras**** - Given a locally compact group G , one can construct the ****group C*-algebra**** $C^*(G)$, which encodes the unitary representations of G . - There are two versions: the ****full**** $C^*(G)$ and the ****reduced**** $C_r^*(G)$ group C*-algebra, depending on whether one considers all unitary representations or only those that appear in the left regular representation.

2. ****Representation Theory and Operator Algebras**** - Unitary representations of G correspond to ****representations of $C^*(G)$ on Hilbert spaces**. - The study of representations of $C^*(G)$ gives insight into the harmonic analysis on G , particularly for non-abelian groups.

3. ****The von Neumann Algebra Connection**** - The ****group von Neumann algebra**** $W^*(G)$ (generated by the left regular representation) is a von Neumann algebra associated with G . - It plays a crucial role in ergodic theory, representation theory, and quantum physics.

4. ****The Plancherel Theorem and Fourier Analysis on Groups**** - For abelian G , $C^*(G)$ is essentially the algebra of Fourier transforms on G . - For non-abelian G , the representation theory of $C^*(G)$ generalizes Fourier analysis.

****2. Study Path**** To systematically approach this topic, follow these steps:

****Step 1: Prerequisites**** - ****Functional Analysis**** - Banach and Hilbert spaces, bounded operators. - The Spectral Theorem. - Gelfand representation of commutative C*-algebras.

- Measure Theory Topology: - Haar measure and integration on locally compact groups. - Borel functional calculus.

- ****Basic Group Theory Representation Theory**** - Unitary representations, irreducible representations. - Induced representations.

****Step 2: C*-Algebras and Their Representations**** - Definition and examples of C*-algebras. - Representations of C*-algebras on Hilbert spaces. - The GNS (Gelfand–Naimark–Segal) construction. ****Step 3: Locally Compact Groups and Their Unitary Representations**** - Basic properties of locally compact groups. - The Peter-Weyl theorem for compact groups. - Induced representations (Mackey theory).

****Step 4: Group C*-Algebras**** - Definition and construction of $C^*(G)$. - Full vs. reduced group C*-algebras. - The role of the left regular representation and the reduced group C*-algebra.

****Step 5: Fourier Analysis on Groups and Plancherel Theorem**** - The dual space of a locally compact group. - Plancherel measure and decomposition of $L^2(G)$. - The role of C*-algebras in noncommutative harmonic analysis.

****Step 6: Advanced Topics and Applications**** - The Baum-Connes conjecture. - Noncommutative geometry and its connection to group C*-algebras. -

Representation theory of Lie groups and its interplay with operator algebras.

— **3. Suggested References** 1. **C*-Algebras:** - J. Dixmier, *C*-Algebras and Their Representations*. - Gert K. Pedersen, *Analysis Now* (for a functional analytic perspective).

2. **Representation Theory Locally Compact Groups:** - N. Higson J. Roe, *Analytic K-Homology*. - Folland, *A Course in Abstract Harmonic Analysis*.

3. **Noncommutative Harmonic Analysis Advanced Topics:** - A. Connes, *Noncommutative Geometry*. - Alain Valette, *Introduction to the Baum-Connes Conjecture*.

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This plan will guide you from foundational knowledge to research-level understanding. Let me know if you need further clarifications or more tailored recommendations!